

COSC 594/650

Nesterov Acceleration

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First-order oracle model and complexity

Function class $\mathcal{F}(D, L)$: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (convex)

Initial gap: $\|x_0 - x^\star\| \leq D$

Lipschitz gradient: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$

■ ORACLE MODEL Ω OF FIRST-ORDER ALGORITHMS

$$x_{k+1} = \sum_{i=0}^k (\alpha_{i,k} x_i + \beta_{i,k} \nabla f(x_i))$$

★ $\alpha_k = (\alpha_{0,k}, \alpha_{1,k}, \dots, \alpha_{k,k})$, $\beta_k = (\beta_{0,k}, \beta_{1,k}, \dots, \beta_{k,k})$ – weights

■ COMPLEXITY OF AN ALGORITHM $\mathcal{A} \in \Omega$

$$\mathcal{C}(\mathcal{A}, \epsilon) = \sup_{f \in \mathcal{F}(D, L)} \min\{k \mid f(x_k) - f(x^\star) \leq \epsilon\}$$

Optimal first-order methods

Optimal complexity: $\inf_{\mathcal{A} \in \Omega} \mathcal{C}(\mathcal{A}, \epsilon)$

Iteration complexity:

$$\sqrt{\frac{3}{32}} \min \left(n, \sqrt{\frac{LD^2}{\epsilon}} \right) \leq \inf_{\mathcal{A} \in \Omega} \mathcal{C}(\mathcal{A}, \epsilon) \leq \sqrt{\frac{4LD^2}{\epsilon}}$$

Thm 3.14, Bubeck Nesterov 1983

Gradient descent is not order optimal

Intuition for Nesterov acceleration

Gradient descent: $x_{k+1} = x_k - \alpha \nabla f(x_k)$

Equivalent minimization: $x_{k+1} = \operatorname{argmin}_x f(x_k) + \langle \nabla f(x_k), x - x_k \rangle + \frac{1}{2\alpha} \|x - x_k\|^2$

Two testing sequences: $\{x_k\}, \{y_k\}$

■ HISTORY-BASED MINIMIZATION

$$A_k f(x_k) \leq \min_y \sum_{i=0}^k a_i (f(y_i) + \langle \nabla f(y_i), y - y_i \rangle) + \frac{L}{2} \|y - x_0\|^2$$

★ $A_k = \sum_{i=0}^k a_i, a_i \geq 0$ – weights

★ $z_k = x_0 - \frac{1}{L} \sum_{i=0}^k a_i \nabla f(y_i)$ – minimizer

Intuition for Nesterov acceleration

Cost-to-go estimate:

$$f(x_k) - f(x^\star) \leq \frac{LD^2}{2A_k}$$

■ HISTORY-BASED MINIMIZATION

$$A_k f(x_k) \leq \min_y \sum_{i=0}^k a_i (f(y_i) + \langle \nabla f(y_i), y - y_i \rangle) + \frac{L}{2} \|y - x_0\|^2$$

★ simplification with $y = x^\star$ yields

$$A_k f(x_k) \leq f(x^\star) \sum_{i=0}^k a_i + \frac{L}{2} \|x^\star - x_0\|^2$$

$$\implies f(x_k) - f(x^\star) \leq \frac{LD^2}{2A_k}$$

Nesterov acceleration

Two testing sequences: $\{x_k\}$, $\{y_k\}$

■ HISTORY-BASED MINIMIZATION

$$A_k f(x_k) \leq \min_y \sum_{i=0}^k a_i (f(y_i) + \langle \nabla f(y_i), y - y_i \rangle) + \frac{L}{2} \|y - x_0\|^2$$

$$\star \textcolor{red}{z}_k = x_0 - \frac{1}{L} \sum_{i=0}^k a_i \nabla f(y_i) - \text{minimizer}$$

■ NESTEROV ACCELERATION

$$\begin{aligned} y_{k+1} &= (1 - \theta_k) x_k + \theta_k \textcolor{red}{z}_k \\ x_{k+1} &= y_{k+1} - \frac{1}{L} \nabla f(y_{k+1}) \end{aligned}$$

$$\star \theta_k = \frac{a_{k+1}}{A_{k+1}}, a_0 \in [0, 1], a_k^2 \leq A_k - \text{parameters}$$

Linear parameter choice

■ LINEAR CHOICE

$$a_k = \frac{k+1}{2}$$

$$\implies A_k = \frac{(k+1)(k+2)}{4}$$

$$\star f(x_k) - f(x^\star) \leq \frac{2LD^2}{(k+1)(k+2)} - \text{accelerated rate}$$

Aggressive parameter choice

■ AGGRESSIVE CHOICE

$$a_k^2 = A_k$$

$$\implies a_k^2 - a_k = a_{k-1}^2, \quad a_k = \frac{1}{2} \left(1 + \sqrt{4a_{k-1}^2 + 1} \right)$$

■ SIMPLE RECURSION

$$\begin{aligned} y_{k+1} &= (1 + t_k)x_k - t_k x_{k-1} \\ x_{k+1} &= y_{k+1} - \frac{1}{L} \nabla f(y_{k+1}) \end{aligned}$$

★ $x_0 = x_1 = 0, a_0 = 0$ – initialization

★ $t_k = \frac{a_k - 1}{a_{k+1}}$ – parameter

Proof of simple recursion (I)

■ NOTATION

$$g_k = \frac{1}{L} \nabla f(y_k), \quad z_k = - \sum_{i=0}^k a_i g_i$$

$$\begin{aligned} y_{k+1} &= (1 + t_k)x_k - t_k x_{k-1} \\ x_{k+1} &= y_{k+1} - g_{k+1} \end{aligned}$$

★ $x_0 = 0, g_0 = 0$ – initialization

■ TELESOPING SUM

$$a_{k+1}y_{k+1} - (a_{k+1} - 1)x_k = a_k y_k - (a_k - 1)x_{k-1} - a_k g_k$$

$$\implies y_{k+1} = (a_{k+1} - 1)(x_k - y_{k+1}) + z_k$$

Proof of simple recursion (II)

■ TELESCOPING SUM (CONTINUED)

$$y_{k+1} = (a_{k+1} - 1)(x_k - y_{k+1}) + z_k$$

$$\implies y_{k+1} = \left(1 - \frac{1}{a_{k+1}}\right) x_k + \frac{1}{a_{k+1}} z_k$$

$$\implies y_{k+1} = (1 - \theta_k) x_k + \theta_k z_k$$

$$\star \theta_k = \frac{a_{k+1}}{A_{k+1}} = \frac{1}{a_{k+1}} - \text{parameter}$$

Iteration complexity of simple recursion (I)

Smoothness & convexity:

$$f(x_{k+1}) - f(x) \leq L \langle g_{k+1}, y_{k+1} - x \rangle - \frac{L}{2} \|g_{k+1}\|^2$$

■ SUFFICIENT DECREASE

$$e(x_{k+1}) - e(x_k) \leq L \langle g_{k+1}, y_{k+1} - x_k \rangle - \frac{L}{2} \|g_{k+1}\|^2$$

■ COST-TO-GO ESTIMATE

$$\begin{aligned} e(x_{k+1}) &\leq L \langle g_{k+1}, y_{k+1} - x^\star \rangle - \frac{L}{2} \|g_{k+1}\|^2 \\ &= L \langle g_{k+1}, z_k - x^\star \rangle + L(a_{k+1} - 1) \langle g_{k+1}, x_k - y_{k+1} \rangle - \frac{L}{2} \|g_{k+1}\|^2 \end{aligned}$$

$$\star y_{k+1} = \left(1 - \frac{1}{a_{k+1}}\right) x_k + \frac{1}{a_{k+1}} z_k$$

Iteration complexity of simple recursion (II)

■ ADDING SUFFICIENT DECREASE TO COST-TO-GO ESTIMATE

$$(a_{k+1} - 1)(e(x_{k+1}) - e(x_k)) \leq (a_{k+1} - 1)L\langle g_{k+1}, y_{k+1} - x_k \rangle - \frac{L(a_{k+1} - 1)}{2}\|g_{k+1}\|^2$$
$$\oplus$$

$$e(x_{k+1}) \leq L\langle g_{k+1}, z_k - x^* \rangle + L(a_{k+1} - 1)\langle g_{k+1}, x_k - y_{k+1} \rangle - \frac{L}{2}\|g_{k+1}\|^2$$

$$\implies a_{k+1}e(x_{k+1}) - (a_{k+1} - 1)e(x_k) \leq L\langle g_{k+1}, z_k - x^* \rangle - \frac{La_{k+1}}{2}\|g_{k+1}\|^2$$

■ REARRANGEMENT

$$\begin{aligned} a_{k+1}^2 e(x_{k+1}) - a_{k+1}(a_{k+1} - 1)e(x_k) &\leq La_{k+1}\langle g_{k+1}, z_k - x^* \rangle - \frac{La_{k+1}^2}{2}\|g_{k+1}\|^2 \\ &= -La_{k+1}\langle g_{k+1}, x^* \rangle - \frac{L}{2}(\|z_{k+1}\|^2 - \|z_k\|^2) \end{aligned}$$

$$\star \quad z_{k+1} = z_k - a_{k+1}g_{k+1}$$

Iteration complexity of simple recursion (III)

$$a_{k+1}^2 e(x_{k+1}) - \underbrace{a_{k+1}(a_{k+1} - 1)}_{= a_k^2} e(x_k) \leq La_{k+1} \langle g_{k+1}, x^* \rangle - \frac{L}{2} (\|z_{k+1}\|^2 - \|z_k\|^2)$$

■ TELESCOPING SUM

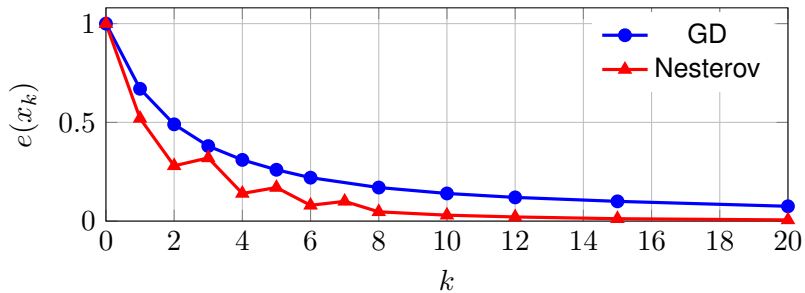
$$\begin{aligned} a_{k+1}^2 e(x_{k+1}) - a_0^2 e(x_0) &\leq L \langle z_{k+1}, x^* \rangle - \frac{L}{2} \|z_{k+1}\|^2 \\ &\leq \frac{L}{2} \|x^*\|^2 \end{aligned}$$

$$\star \quad z_{k+1} = - \sum_{i=0}^{k+1} a_i g_i, \quad z_1 = 0$$

Iteration complexity:

$$e(x_{k+1}) \leq \frac{LD^2}{2A_{k+1}} \leq \frac{2LD^2}{(k+2)^2}$$

Acceleration is generally non-monotone



Optimal first-order methods for strongly convex functions

Iteration complexity of GD: $\mathcal{O}\left(\kappa \log \frac{1}{\epsilon}\right)$

QUESTION: Can we make Nesterov's method also order-optimal?

Restart Nesterov's method every $T = \sqrt{C_1 \kappa}$ iterations

$$f(x_{iT}) - f(x^*) \leq C_2 \frac{L \|x_{(i-1)T} - x^*\|^2}{T^2} \leq \frac{2C_2}{C_1} (f(x_{(i-1)T}) - f(x^*))$$

$$\implies f(x_k) - f(x^*) \leq \left(\frac{2C_2}{C_1}\right)^{k/T} (f(x_0) - f(x^*))$$

Iteration complexity of restarted Nesterov: $\mathcal{O}\left(\sqrt{\kappa} \log \frac{1}{\epsilon}\right)$

Convergence analysis of optimization algorithms

■ ASYMPTOTIC CONVERGENCE

■ RATE OF CONVERGENCE

■ ITERATION COMPLEXITY ANALYSIS

Method	Convex	Strongly convex
GD	$O(1/\epsilon)$	$O(\kappa \log 1/\epsilon)$
Nesterov	$O(1/\sqrt{\epsilon})$	$O(\sqrt{\kappa} \log 1/\epsilon)$

Questions?

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